Ashwin Chembu

(341) 356-1722 | ashchembu@gmail.com | San Francisco, California

PROFILE

Frontend engineer building responsive React and TypeScript interfaces, collaborative FigJam tools, and interactive Three.js experiences. Connects reusable components with product workflows, authorization, and testing.

TECHNICAL SKILLS

Languages / UI: TypeScript, JavaScript, HTML, CSS, React, Next.js, Angular, Vue, Tailwind CSS
Interaction: Three.js, Figma API, WebSockets, Capacitor, React Router, responsive components
Testing / Delivery: Vitest, React Testing Library, Playwright, Firebase emulators, Git, Vite

EXPERIENCE

Business Technology Solutions Associate | ZS Associates **South San Francisco, CA | Jun 2024-Present**

- Implemented **React**/JavaScript account-plan submission and manager approval with CRM-backed data services, timestamps, review history, and confirmation paths, simplifying field-user actions while preserving review traceability.
- Refined **React** CRM launchers and account-profile screens with sorting, access filtering, responsive columns, constrained percentage inputs, and edit/save flows, aligning user guidance with the implemented navigation and data mappings.
- Added cached-section recovery and explicit retries to **React** CRM views, preserving last-good data after refresh failures and ignoring obsolete requests; **five** regression tests covered initial errors, retries, and selected-plan changes.
- Translated ambiguous healthcare stakeholder needs into testable rules, API integrations, and source-to-target mappings, documenting acceptance criteria and unresolved decisions so delivery teams could implement scoped changes.
- Converted CRM metadata, live workflow observations, meeting transcripts, and stakeholder comments into user guides, decision logs, executive decks, and prioritized backlogs that connected implementation status to explicit handoffs.

Founder and Lead Developer | Tallyrus **Jun 2023-Sep 2025**

- Designed **React** rubric and essay-review interfaces that connect teacher materials to generated feedback, incorporating educator testing and iteration so users could inspect writing quality while retaining consistent scoring logic.
- Integrated **React**/**TypeScript** and Express product workflows with Google OAuth, protected routes, MongoDB persistence, and secure sessions, connecting authenticated educator access to document and essay-feedback operations.
- Led product planning, educator testing, and iterative development for the collaborative essay-feedback platform, translating user feedback into rubric and review workflows and raising **five figures** to fund product development.
- Connected Express/Multer upload routes for teacher files, rubrics, and historical essays to AWS S3 storage and MongoDB assignment records, making educator materials available to downstream grading and review workflows.
- Implemented a Flask grading API that validates rubric, essay, prompt, and historical-example inputs before generating JSON feedback, connecting model requests to the review workflow without treating automated output as a final grade.

Data Analyst / Web Development Lead | Tech4Good at UCSC **Mar 2023-Jun 2025**

- Built **TypeScript**/**Figma** API chat, text-entry, polling, and Excel/XLSX export workflows in a collaborator-owned FigJam repository, enabling research and classroom teams to collect, discuss, and export structured participant responses.
- Built a responsive Angular career-exploration component for top-paying industries, implementing distinct desktop, tablet, and phone layouts so students could inspect industry information across screen sizes.
- Implemented anonymous and multi-select polling with **TypeScript**/**Figma** APIs, including changed votes, per-option response displays, grouped widgets, and reusable Likert selections so teams could collect and revisit collaborative feedback.
- Separated chat, text-entry, and polling state across **three MongoDB databases**, using change streams, populated message logs, and persisted voter records to synchronize threads, edits, group responses, and reveal/reset operations.

PROJECTS

3D Commerce Experience | Interactive Storefront **Public UI prototype**

- Built a **React**/**TypeScript** storefront with reusable product cards, navigation, modals, and responsive layouts, connecting product discovery, search, cart, checkout, and account pages into a complete interactive commerce prototype.
- Developed a **Three.js** product configurator with GLB assets and Draco decoding, adding independent garment customization, color, and placement controls so shoppers could inspect and customize a varsity jacket from multiple viewing angles.
- Improved mobile WebGL recovery by preloading poster, GLB, and Draco assets, delaying the canvas until visible pixels rendered, and providing retryable poster fallback after context loss; validated desktop/mobile views and production builds.

Sylk | Multi-Tenant Supply-Chain Platform **Private organization product | 2025-Present**

- Built **React** 19/**TypeScript** supply-chain workflows with Firebase for organization onboarding, line-item shipments, fulfillment, files, and real-time chat, connecting retailer and manufacturer roles through a shared operational product.
- Packaged the full product as a Capacitor Android app, adding native-safe routing, Firebase Authentication, safe-area handling, mobile file picking, and phone-specific navigation across orders, inventory, billing, chat, and team access.
- Tested tenant access, reusable components, and complete supply-chain journeys with Vitest, **React** Testing Library, Firebase emulators, and **Playwright,** checking permission boundaries alongside onboarding and shipment workflows.

Healthcare Dashboard | Workflow UI Prototype **Public UI prototype**

- Designed healthcare account and trial-management dashboard prototypes in **React**/**TypeScript** and Vue, translating account hierarchies, ongoing trials, and site progression into scannable interfaces for stakeholder design review.
- Built reusable metric cards, progress steppers, follow-up queues, and barrier summaries, organizing complex healthcare workflow status into consistent component patterns that made pending actions easier to inspect in product demos.
- Added task-editing and trial-detail modals alongside account hierarchy views, connecting high-level progress summaries to detailed workflow interactions so stakeholders could review the prototype and validate interface behavior.

EDUCATION

University of California, Davis | B.S. Computer Science | GPA: 3.7 **Davis, CA | Graduated Jun 2025**

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu